Does Cost-Efficiency Travel? Raw Cost Portability, Success-Adjusted Efficiency, and Matched Agent Trajectories

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Abstract

Agent leaderboards increasingly report task success and dollar cost, inviting practitioners to select a "cost-efficient" system from a single benchmark. We ask whether this signal transfers across workloads, whether it reflects stable label-level cost propensity, and whether observable agent trajectories expose a scaffold contribution. First, using public Holistic Agent Leaderboard (HAL) data and a fixed displayed Generalist Agent scaffold, we compare repeated displayed model labels across six benchmarks. Raw dollar-cost ranks transfer in high-overlap pairs, but a pooled benchmark-plus-label log-cost model explains 82.0% of observed variation and residual cost ranks do not remain significant. Aggregate dollars per expected success are less portable than raw cost. Leave-one-benchmark-out prediction succeeds on four held-out workloads and fails on two lower-overlap workloads. Second, we analyze a separate public unencrypted matched trace sample: across 977 exact same-task/same-model pairs, OpenHands has lower median turns and tool calls than SWE-agent without a detected resolved-status difference. These studies establish that apparent agent cost efficiency mixes persistent displayed-label propensity, benchmark context, and scaffold trajectory structure. They do not identify tokens, provider prices, or dollar failure costs, which remain unavailable in the public snapshots.

1 Introduction

Agent evaluations increasingly expose a joint accuracy–cost tradeoff. This is a welcome correction to accuracy-only leaderboards: long-horizon systems can invoke tools, grow context, retry actions, and consume substantial resources. Yet a common practical inference remains underexamined: a label that is cheap on one benchmark is presumed to be an efficient deployment choice elsewhere. That inference conflates at least three mechanisms: persistent provider/model/configuration cost propensity, benchmark-specific demands, and scaffold execution behavior.

A concrete ranking example illustrates why this distinction matters. In the GAIA–SWE-mini pair, 17 labels overlap and raw dollar-cost rank correlation is 0.89. Yet their aggregate dollars-per-expected-success rank correlation is 0.29, and the paired raw-minus-CPS contrast is 0.60 with a 95% configuration-resampling sensitivity interval [0.14, 1.13]. Thus, a practitioner using a GAIA-like leaderboard to rank raw spend for a SWE-like workload might obtain a useful prior, while a practitioner needing a ranking of expected dollars per successful task would not. This is an aggregate ranking illustration, not an assertion about named models or production outcomes.

A second, explicitly separate trace study supplies a mechanism-oriented complement. The HAL public trace archives are encrypted, so they cannot answer how turns or tool calls differ. We therefore use a public unencrypted Open-SWE-Traces sample that permits exact task/model matching across two scaffolds. This trace extension measures structured trajectory burden proxies, not tokens or dollars.

Our contributions are:

1. a fixed-scaffold, coverage-aware analysis of raw cost, success-adjusted cost, and frontier portability across six HAL workloads;
2. held-out LOBO prediction and an observational displayed-label cost-propensity adjustment that separates persistent raw-cost structure from residual benchmark-specific structure;
3. a matched external trace extension showing substantial scaffold differences in turns and tool calls at fixed task and model, and showing that the popular "expensive failure" premium dissolves under exact task matching; and
4. a reproducible artifact with source hashes, sensitivity analyses, and explicit boundaries on unavailable token, price, retry, and failure-cost data.

2 Related work

Kapoor et al. [2026] introduce HAL as cost-aware infrastructure across coding, web, science, and customer-service tasks. Kapoor et al. [2024] motivate joint accuracy–cost reporting and convex-hull frontiers. Ndzomga [2026] ask whether task subsets preserve ranks within agent benchmarks. Benchmark^2 studies cross-benchmark rank consistency for static LLM benchmarks. SWE-Effi examines resource effectiveness within software engineering, while Open-SWE-Traces releases public agent trajectories [Fan et al., 2025, Ahmad et al., 2026]. Our contribution is a coverage-aware test of whether a public cost signal predicts a held-out interactive-agent workload, whether that signal survives success adjustment, and what matched traces can reveal about scaffold burden.

3 Primary HAL study

3.1 Data and cohort

We use the public `all_leaderboards_costs_HAL.csv` distributed with Ndzomga [2026]. The frozen snapshot contains 242 displayed evaluations across nine benchmarks, 11 displayed scaffolds, and 29 displayed model labels. Most evaluations are single-run. Dollar cost and accuracy are available, but the CSV has no model/provider alias, prompt/output/cache token components, API-call count, tool-call count, retry count, or per-task trajectory records.

The only scaffold repeated broadly enough for the primary analysis is `HAL Generalist Agent`. We retain six workloads with adequate repeated-label overlap: CORE-Bench Hard, GAIA, SciCode, ScienceAgentBench, SWE-bench Verified Mini, and TAU-bench Airline. This yields 86 rows and 25 displayed labels. USACO has only one Generalist observation and is excluded. We match only exact public `Models` strings within the exact public `Agent Name`; this is not a controlled base-model identity claim.

3.2 Metrics

For every benchmark pair with at least five shared labels, we compare ranks using Spearman ρ and Kendall τ_b, reporting overlap N. Pairwise rank intervals use 10,000 configuration-resampling draws. Raw-cost Holm–Bonferroni correction applies to 15 tests; CPS uses its own 20,000-draw permutation tests and Holm correction across 15 CPS tests. These quantify finite-label sensitivity, not rollout uncertainty.

We define aggregate dollars per expected success as

$$\mathrm{CPS}_{i,b}(\epsilon) = \frac{C_{i,b}}{\max(A_{i,b}, \epsilon)},$$

Table 1: High-overlap raw dollar-cost rank transfer and residual cost-rank sensitivity. Residual ranks remove observed displayed-label and benchmark log-cost effects; they are not list-price-adjusted costs.

Pair	N	Raw ρ	Residual ρ	Perm. p
CORE–GAIA	14	0.93	0.23	0.426
CORE–SWE-mini	14	0.74	-0.49	0.079
CORE–TAU	12	0.78	-0.03	0.921
GAIA–SWE-mini	17	0.89	0.03	0.917
GAIA–TAU	14	0.86	0.13	0.648
SWE-mini–TAU	14	0.82	-0.16	0.570

where $C_{i,b}$ is displayed aggregate dollar cost and $A_{i,b} \in [0, 1]$ is accuracy. The primary floor is $\epsilon = 0.01$; $\epsilon = 0.001$, $\epsilon = 0.05$, and zero-accuracy exclusion are reported as sensitivities. CPS is an aggregate ratio, not mean spending among successful trajectories.

We compute standard discrete nondominated frontiers and a HAL-inspired origin-anchored convex-hull reconstruction. A 5% tolerance sensitivity requires both 5% lower cost and 5% higher accuracy for dominance. It is a near-tie stress test, not a measurement-error model.

3.3 Cost-propensity and held-out models

To assess whether raw cost transfer is dominated by stable observed label propensity, we fit

$$\log C_{i,b} = \mu + \alpha_i + \beta_b + \epsilon_{i,b},$$

where α_i is a displayed-label fixed effect and β_b is a benchmark effect. This is an observational adjustment, not a provider-list-price adjustment. Stable label propensity can include prices, reasoning settings, routing, token behavior, cache treatment, and unrecorded setup.

For leave-one-benchmark-out (LOBO) prediction, we hold out a benchmark, center log cost within each training benchmark, average each label's centered cost over the remaining benchmarks, and predict the held-out rank. Labels must occur in at least two training workloads. We assess each held-out rank correlation with a 20,000-draw within-benchmark label-pairing permutation null.

4 Primary results

4.1 Raw dollar-cost rank transfers, but residual cost rank does not

Across the six pairs with at least 12 shared labels, raw dollar-cost Spearman correlation ranges from 0.74 to 0.93, with mean 0.84. These associations survive the pre-specified raw-cost multiplicity and permutation checks. The pooled cost-propensity model has 30 parameters on 86 rows, leaving 56 residual degrees of freedom; its $R^2 = 82.0\%$ (adjusted $R^2 = 72.7\%$) is descriptive in-sample evidence rather than the primary portability test. After removing benchmark and displayed-label cost propensity, no high-overlap residual-cost rank association is significant under a 20,000-draw permutation test. LOBO prediction below supplies the held-out evidence.

4.2 Success adjustment is less portable

Across the same six high-overlap pairs, the CPS mean is 0.59 under the 1% floor, compared with 0.84 for raw cost; this average is descriptive. CPS association survives 20,000-draw permutation testing and Holm correction only for CORE–GAIA ($\rho = 0.87$, 95% sensitivity interval [0.63, 0.96]), CORE–TAU (0.83,

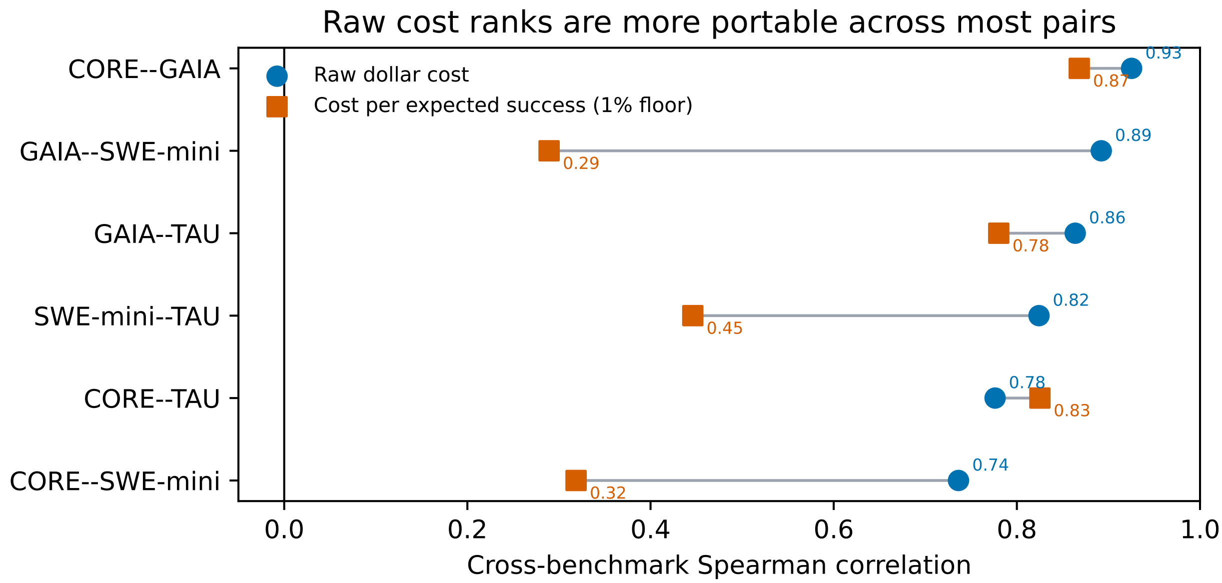

Figure 1: Raw-cost versus CPS rank transfer for high-overlap pairs. CPS uses the primary 1% denominator floor.

[0.31, 1.00]), and GAIA–TAU (0.78, [0.42, 0.94]). The paired raw-minus-CPS difference is clearly positive only for GAIA–SWE-mini (0.60, 95% interval [0.14, 1.13]). Other pairwise CPS estimates remain uncertain. Thus a label that is predictably cheap is not necessarily predictably cheap per expected success on a different workload.

4.3 LOBO identifies portable and nonportable raw-cost settings

In LOLO influence checks, the positive LOBO findings are not single-label artifacts: CORE remains $\rho \in [0.73, 0.91]$, GAIA [0.84,0.92], SWE-mini [0.88,0.92], and TAU [0.79,0.95] across every omission. SciCode is unstable ([-0.50,0.14]); ScienceAgentBench remains negative ([-0.77,-0.43]) but its original $N = 7$ test is non-significant. SciCode's mean accuracy is 2.1% with 33.3% zero-accuracy labels; ScienceAgentBench has the narrowest log-cost dispersion. These six-benchmark diagnostics are descriptive and do not identify a domain effect.

4.4 Frontier membership is non-universal

No label tested on at least five workloads appears on every observed accuracy–cost frontier under discrete nondominance, the HAL-inspired hull, or 5% tolerance. The stronger claim that no label appears on more than 4/6 frontiers is not robust: under tolerance, o4-mini Low appears on 5/6. We therefore retain only the non-universal-winner conclusion.

5 External matched trajectory study

5.1 Data and design

The HAL aggregate CSV cannot identify execution mechanisms because its traces are encrypted. We therefore conduct an external, separate matched trace analysis using public unencrypted Open-SWE-Traces data.

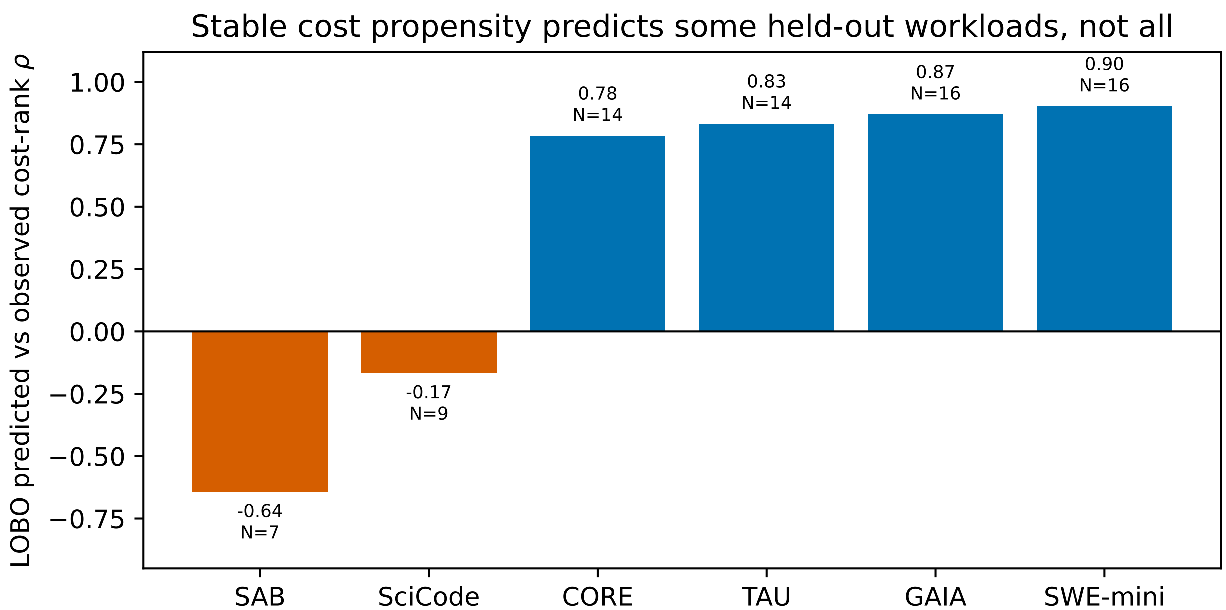

Figure 2: LOBO predicted versus observed raw-cost ranks. Blue bars differ from the 20,000-draw label-pairing permutation null at 0.05; orange bars do not.

We select a documented four-shard boundary sample for MiniMax-M2.5 under OpenHands and SWE-agent. The boundary sample is reproducible but not a random sample of the full corpus. We exact-match task ID and model, exclude duplicate task/model/scaffold records, and obtain 977 pairs.

The trace schema contains resolved status, message roles, explicit tool calls, message text, and reasoning text. It does not contain billed tokens, dollars, provider usage, retries, or latency. We measure trajectory turns, assistant turns, explicit tool calls, tool-result turns, total message characters, and reasoning-message characters. Character counts are tokenizer-free text-volume proxies, not tokens or costs. Paired comparisons use two-sided Wilcoxon signed-rank tests.

5.2 Scaffold changes trajectory burden at fixed task and model

OpenHands has lower matched medians: 123 versus 149 trajectory turns, 61 versus 74 assistant turns/tool calls, and 176,708 versus 194,263 total trajectory characters. The OpenHands/SWE-agent median ratio is 0.85 for turns (10,000-bootstrap interval [0.82,0.87]) and 0.85 for tool calls ([0.82,0.86]); OpenHands is lower on 69.7% of pairs for each. Characters have ratio 0.95 ([0.92,0.98]) and are lower on 55.1% of pairs. Resolved status has no detected paired difference ($p = 0.197$). This supports a scaffold difference in observable trajectory structure, not lower tokens, dollar cost, or a causal explanation of HAL rankings.

5.3 Failure carries no burden premium at fixed task

The same sample tests a popular cost heuristic: comparing each scaffold's failed versus resolved trajectories across tasks, failures look 19–36% heavier (turns ratio 1.21 [1.07,1.36] OpenHands, 1.19 [1.07,1.35] SWE-agent; trajectory characters up to 1.33), the "expensive failure" premium reported in prior resource-effectiveness work. Exact task matching dissolves it. Among the 107 discordant pairs — identical task, identical model, one scaffold resolved, the other failed — the failed side is heavier only 47–52 times out of 107 on every metric (two-sided binomial $p \geq 0.25$), and after normalizing each trajectory by its own

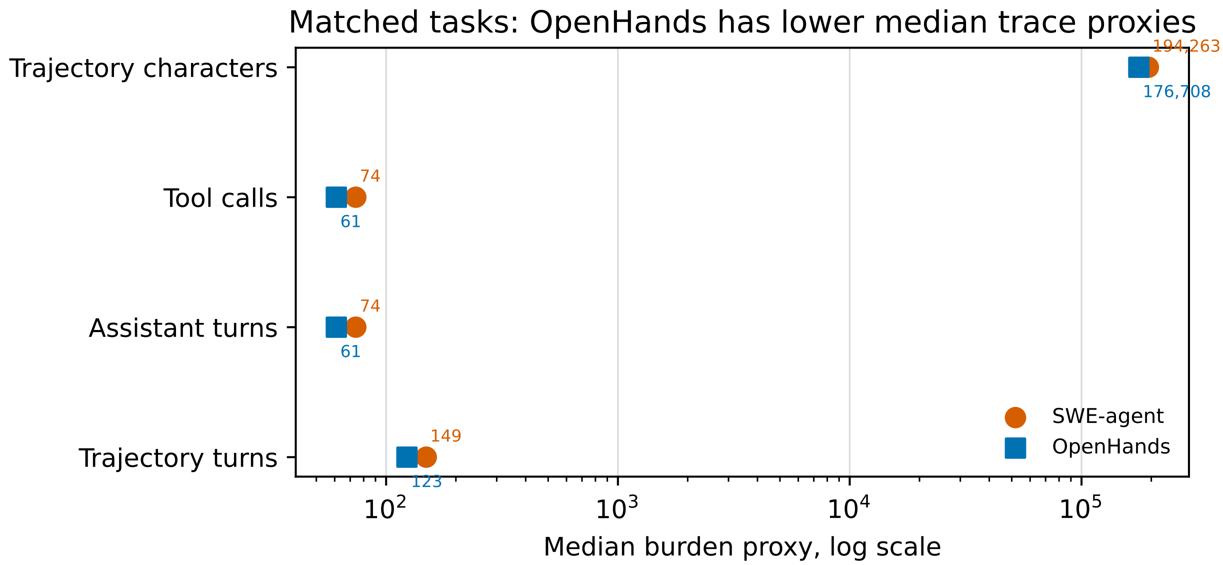

Figure 3: Matched MiniMax-M2.5 task pairs in the Open-SWE boundary sample ($N = 977$). Markers show scaffold medians on a log scale for trace proxies, not tokens or dollars.

scaffold's median burden, the failed/resolved median ratio is 0.98–1.06 with sensitivity intervals spanning 1 throughout. Hard tasks make agents both burn more and fail; unconditioned failure-cost ratios therefore measure workload composition, not a property of failing runs. Failure-rate-times-premium accounting inherits exactly the composition problem documented for raw aggregate dollars in Section 4.

6 Discussion

The combined results clarify what a cost-aware agent leaderboard can and cannot establish. Raw dollar-cost rankings are predictive for some held-out workloads, but their portability is largely associated with stable observed displayed-label cost propensity. CPS portability is weaker, so raw cheapness is not interchangeable with deployment efficiency. Independent matched traces further show that scaffolds change measurable execution structure, even when task and model are fixed.

This is a secondary-data analysis with clear construct boundaries. The primary CSV has sparse, nonrandom coverage, mostly single-run rows, and public display labels that omit prompts, tools, budgets, and routing. Cost-propensity adjustment is not provider-price control. CPS is not successful-run cost. The external trace study is not merged with HAL, uses a nonrandom boundary sample, and uses burden proxies rather than billing fields. We cannot compute failure-cost ratios, tail dollar costs, tokens, cache usage, retries, or a true three-dimensional accuracy–cost–reliability frontier from public data. These limitations motivate public release of aggregate usage and outcome telemetry with leaderboard results; Appendix A probes how far two public unencrypted telemetry corpora go toward that standard.

7 Conclusion

Observed dollar-cost portability is real in several high-overlap agent benchmark pairs, but it should not be read as a universal deployment-efficiency signal. It is partly persistent label-level cost propensity, weakens after success adjustment, and coexists with substantial scaffold differences in matched trajectory structure.

Agent leaderboards should report not only accuracy and raw dollar cost, but also success-adjusted outcomes and trace-level usage fields needed to distinguish price, scaffold, and execution mechanisms.

A Cross-source telemetry anatomy

The primary CSV reports dollars without usage components, and the HAL trace archives are encrypted. To probe how much anatomy public unencrypted telemetry can supply, we audited two independent coding-agent telemetry corpora. TraceLab contributes 8,058 real-world sessions comprising 665,453 LLM rounds and 743,819 tool calls; the MIMO Claude Code release contributes 1,017 sessions (4,690 rounds) from a single model label. Neither source exposes benchmark task outcomes or dollar billing, so all summaries are observational.

Table 2 reports the TraceLab anatomy. Median rounds carry 132,092 input tokens against only 249 output tokens, with a 93.7% mean cache-read share; median final-round context is $3.6\times$ the initial round (P90 $9.9\times$). Resource burden is extremely heavy tailed: the top 1% of sessions hold 48.3% of summed input-token volume and the top 10% hold 88.2%. Tool errors also cluster: after a successful tool call the next call errs 4.9% of the time, versus 21.0% after an erroneous call, a $4.29\times$ conditional increase. Retrying is the norm: 88.3% of errored calls are followed by another call to the same tool in the same session, and 77.9% of those immediate retries succeed. The failure tail is nevertheless pathological — consecutive-error runs have median length 1 but P90 length 3 and a maximum of 368. Session burden rises steeply with observed error count, from a median of 0.34M summed input tokens at zero errors to 14.65M at six or more (Figure 4).

Table 2: TraceLab key telemetry metrics.

Metric	Value
TraceLab rounds	665,453
TraceLab sessions	8,058
Median input / output tokens per round	132,092 / 249
Mean cache-read share	93.7%
Median context growth per session	47,267 tokens
Top 1% input-token share	48.3%
Next tool error after success / error	4.9% / 21.0%
MIMO sessions / rounds	1,017 / 4,690
MIMO next tool error after success / error	1.0% / 4.3%
Errored calls with a later same-tool retry / retry succeeds	88.3% / 77.9%
Consecutive-error run length, median / P90 / max	1 / 3 / 368

The MIMO corpus replicates the error cascade at an independent site and model: the next tool call errs 4.3% of the time after an error versus 1.0% after a success, a $4.33\times$ conditional increase that closely mirrors the TraceLab ratio, with the same directional burden gradient across error-count bins (median session input 0.40M, 0.94M, and 1.12M tokens at zero, one, and two to five errors). Two audit boundaries apply. First, in every released MIMO session the per-round usage block repeats one constant tuple, so within-session token dynamics cannot be validated in that source; we therefore use it only for counts, totals, cross-session concentration (top 10% of sessions hold 37.9% of summed input), and content-derived error sequences. Second, neither source links usage to task success or billing, so these are composition and association findings, not failure-cost measurements.

Table 3 summarizes what each audited source can and cannot answer. TraceLab exposes token/cache/tool anatomy without outcomes; Open-SWE supplies exact task/model/scaffold matching with resolved status but no token or billing fields; HAL supplies outcome-linked dollars but encrypts traces. No public source

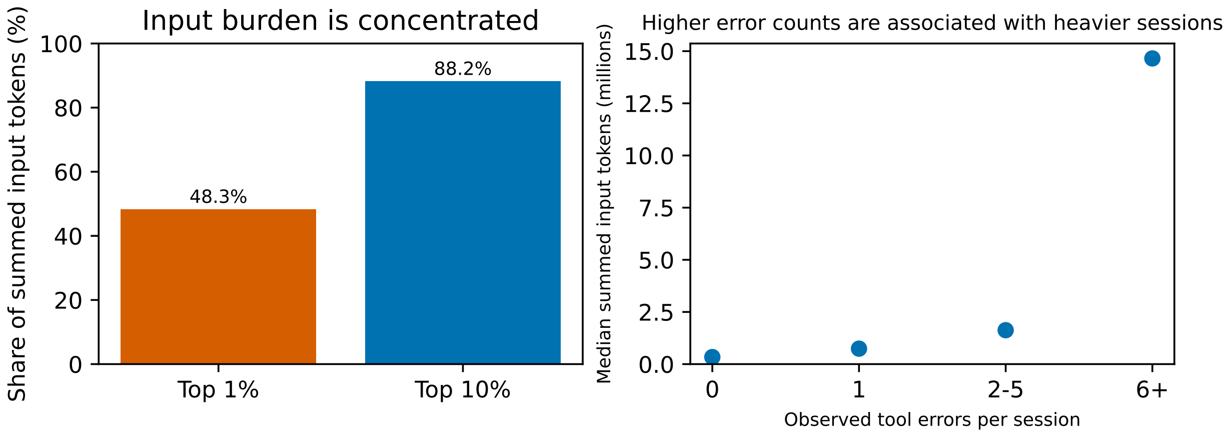

Figure 4: TraceLab tail concentration and the observational association between observed tool-error counts and session input burden.

Table 3: Cross-source telemetry anatomy and scope boundaries.

Source	Sessions / rounds	Context/cache result	Scope
TraceLab	8,058 / 665,453	93.7% mean cache-read share; median last/first context 3.6x	Real sessions; no task outcome or dollars
MIMO Claude Code	1,017 / 4,690	753M cache-read vs 46M uncached input; error cascade 4.33x after a tool error	One model; per-round usage constant within session; outcome unverified
Open-SWE	977 matched pairs	OpenHands/SWE-agent tool-call ratio 0.85 [0.82, 0.86]	Exact task/model match; no tokens or dollars

we audited jointly exposes task, model, scaffold, outcome, token/cache use, retries, latency, and billing. The replicated anatomy makes the missing joint release concrete rather than hypothetical: context-carrying dominates inference volume, tails dominate aggregate burden, and tool-error spirals mark expensive sessions, but none of these can currently be attached to a leaderboard row.

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